Emmanuel McGrail

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Education

University of Pittsburgh Pittsburgh, PA
B.S in Data Science, B.S in Computer Science, Minor in French *GPA: 3.76 | Spring 2027*
Relevant Coursework: Data Structures & Algorithms, Algorithm Implementation, Intro to Systems Software, Computer Organization & Assembly Language, Intro to Computer Vision, Intro to Deep Learning

École Nationale Supérieure de l'Électronique (ENSEA) Cergy, France
Engineering Study Abroad *Spring 2026*
Relevant Coursework: Operating Systems, Signals & Systems, AI & Big Data, Microprocessors

Technical Skills

Technical Languages: Python, C++, Java, R, C#, C, SQL (MySQL)
Communicative Languages: English, French
Libraries/Tools: PyTorch, Nvidia CUDA, ONNX Runtime, Transformers, scikit-learn, scipy, Pandas, NumPy, Spring, Apache Kafka, Apache Spark, Snowflake, AWS Tools (Lambda, Step Functions, Athena, S3)
Awards/Honors: Dean's List, French Dedication Award, EU Economic Project Award, All-Academic Athlete

Professional Experience

Bank of New York (BNY) Pittsburgh, PA
Engineering Intern *May 2025 – Aug. 2025, May 2026 – Present*
- Leading a team of interns to design and build an SLA monitoring dashboard covering thousands of daily extract/feed file deliveries, backed by a configurable SLA rules engine and API, selected to head the effort based on system expertise and prior tenure on the team
- Engineered a data reprocessing algorithm through Spring and Kafka to automatically resolve failed enrichment pipeline messages, increasing system reliability
- Optimized reconciliation workflows, resulting in $1.7M cost savings and an 8x reduction in processing time
- Migrated legacy data zones into a strategic architecture, improving cross-functional access for risk analysis teams

University of Pittsburgh Athletics Pittsburgh, PA
Data Engineering Intern *Sept. 2024 – Dec. 2026*
- Developed a proprietary data pipeline for the diving team using AWS Lambda, processing and grading performance data for thousands of divers to track Pitt athletes' improvement over time
- Enhanced recruitment workflows by integrating position-specific metrics into football pipelines and creating PowerBI dashboards for donors alongside business teams
- Engineered and deployed an LLM-powered TEXT2SQL tool, simplifying SQL query generation and democratizing data access for non-technical team members

Projects

fastdist | *C++, CUDA, PyBind11, Python* Dec. 2025 – Present
- Developed a high-performance mathematics library in C++ and CUDA to accelerate statistical computation and batching operations in Python
- Achieved ~2.5x computational speedup compared to standard libraries by optimizing GPU memory management and kernel execution

STEVE & Minecraft Data API | *Java, Python, PyTorch, ONNX, SpigotAPI* June 2025 – Present
- Updated an old personal project *GenerativeTerrain* into STEVE (Spatial Terrain Engineering & Voxel Embedding), overhauling the plugin API and ONNX inference pipeline to generate terrain in under 3 seconds per chunk
- Built an in-game data collection API (custom commands to extract and label live chunk data) feeding a pipeline that processes roughly one chunk per second for training data generation
- Deprecated legacy CNN/transformer training in favor of a custom Voxel Neural Network, paired with a configurable post-processing phase for procedural ore and resource placement
- Released the pre-trained model (Hugging Face) and chunk dataset (Hugging Face) for public, open-source use

PittAPI | *Python, Requests, BeautifulSoup, UnitTests* May 2024 – Present
- Contribute to PittAPI, an open-source Python library (100+ GitHub stars, published on PyPI) that provides programmatic access to University of Pittsburgh course, dining, library, and shuttle data
- Designed and shipped full study room reservation support end-to-end, including booking creation, availability lookups, and cancellations, as a new core module used by student-facing applications